

Mechanistic Interpretability in Small CNNs via Privileged Primitive Bases on Synthetic Shapes

Heri

Abstract

Mechanistic interpretability aims to recover *faithful, intelligible* descriptions of the computations implemented by trained neural networks. We propose a controlled vision testbed for mechanistic analysis in which the data-generating process is explicitly factorized into simple geometric primitives (e.g., line, circle, triangle, square) and nuisance transformations (translation, rotation, scale, stroke width, and additive noise). Within this setting, we compare (i) *basis-constrained* convolutional models whose first-layer kernels are partially or fully fixed to human-designed templates (a privileged basis), and (ii) *compact learned* models with similar parameter counts but unconstrained filters. Because the relevant features and symmetries are known by construction, we can evaluate interpretability claims against ground truth using multiple complementary probes: feature visualization and inversion to expose learned detectors and invariances [9, 10, 14]; attribution and localization methods to relate predictions to pixels and intermediate maps [8, 11, 12]; and unit-level semantic scoring to quantify alignment between internal channels and labeled concepts [1]. Finally, we frame hypothesis testing in explicitly interventional terms, drawing on formal accounts of mechanistic explanation and tools for rigorous circuit evaluation [4, 6, 7]. The resulting benchmark is intended to support reproducible, quantitative comparisons between architectural inductive bias, representational structure, and mechanistic interpretability outcomes.

1 Introduction

“All models are wrong, but some are useful.” [2]

Convolutional neural networks (CNNs) provide a natural arena for mechanistic interpretability because their computations are spatially structured, compositional across layers, and often amenable to localized visualization and intervention. Nonetheless, standard interpretability practice in vision frequently relies on qualitative inspection or weakly specified hypotheses about what internal units represent. This makes it difficult to distinguish *diagnostic* explanations (correlated with outputs) from *causal* explanations (necessary for the computation), and to compare different inductive biases in a principled way.

This paper outlines and motivates a minimal, fully controlled benchmark in which both the task and the candidate mechanisms are deliberately simple. We train CNNs to classify synthetic black-and-white images containing one of several geometric classes (e.g., line, circle, triangle, square), while systematically varying known nuisance factors such as rotation and translation. We then compare two families of models: (i) *basis-constrained* networks whose early convolutional kernels are fixed (or softly constrained) to a predefined dictionary of templates (e.g., oriented line detectors, corner detectors, circular arcs), and (ii) *compact learned* networks of comparable capacity whose kernels are learned from scratch.

The central methodological goal is not to maximize accuracy, but to create conditions where mechanistic claims are *testable*: the data have known generative factors; the models have transparent architectural bottlenecks; and interpretability analyses can be quantitatively evaluated against ground truth. We adopt a multi-method toolkit spanning visualization, attribution, concept testing, semantic unit scoring, and causal interventions [1, 4, 6–12, 14].

Contributions.

1. A controlled dataset specification for geometric classification with explicit nuisance-factor control (rotation/translation/scale/noise) that supports mechanistic evaluation.
2. A comparative study design contrasting privileged-basis (fixed-template) CNNs with similarly small learned CNNs, enabling analysis of how inductive bias shapes internal representations.
3. A quantitative evaluation protocol that triangulates mechanistic hypotheses using (a) visualization/inversion [9, 10, 14], (b) attribution and concept sensitivity [8, 11, 12], (c) unit semantics [1], and (d) interventional tests of explanatory adequacy [4, 6, 7].
4. A discussion of architectural alternatives that build known symmetries into representations (equivariance/invariance) and their implications for interpretability [3, 5, 13].

2 Related Work

2.1 Visualization and inversion in CNNs

Early mechanistic analysis of CNNs emphasized methods to *visualize* what intermediate feature maps respond to. Zeiler and Fergus [14] introduced deconvolutional and related techniques to map activations back into pixel space, making internal detectors more legible. Complementarily, representation inversion methods explicitly solve for inputs that match given internal codes, thereby revealing what information is retained and what invariances are induced by the network [9]. Feature visualization via activation maximization and its refinements systematize this idea by synthesizing inputs that strongly drive specific units or channels, offering a lens on what a unit “looks for” [10]. These tools motivate our use of synthetic shapes: when the true factors of variation are known, one can test whether visualizations and inversions recover the intended generative structure.

2.2 Attribution, localization, and concept-based testing

Gradient-based methods connect outputs to inputs by measuring sensitivity. Grad-CAM produces class-discriminative heatmaps by combining gradients with convolutional feature maps, supporting localization-style explanations for CNN predictions [11]. Integrated Gradients provides an axiomatic approach to feature attribution by integrating gradients along a path from a baseline input to the input of interest, yielding attributions with desirable invariance properties [12]. Moving beyond pixels, concept-based methods define human-interpretable directions in representation space and test the dependence of predictions on those concepts; TCAV operationalizes this by estimating directional derivatives along concept activation vectors [8]. Because our benchmark defines concepts such as “corner,” “arc,” or “orientation” explicitly (via generators and/or fixed kernels), it supports principled evaluations of whether concept tests track the factors that truly drive classification.

2.3 Quantifying unit semantics

A recurring question is whether individual channels correspond to coherent semantic features. Network Dissection proposes a quantitative framework: measure alignment between unit activations and a labeled concept dataset to assign semantic labels and scores to units [1]. Although originally developed for natural-image concepts, the underlying idea (alignment between activation masks and concept masks) transfers cleanly to synthetic domains where concept labels are exact by construction (e.g., edge orientation fields, corner locations, curvature). This enables direct comparisons between basis-constrained and unconstrained models in terms of how many units attain high concept alignment and how sharply those alignments concentrate.

2.4 Causal testing of mechanistic hypotheses

A mechanistic explanation should be evaluated by interventions, not only correlations. Recent formal work frames mechanistic interpretability via *causal abstraction*: high-level algorithmic descriptions are valid insofar as they preserve interventional behavior under appropriately matched transformations [6]. Causal scrubbing operationalizes this perspective by testing hypothesized circuits through behavior-preserving resampling interventions that ablate purportedly irrelevant computations while retaining relevant ones [4]. Activation patching (and related interchange interventions) provides a practical family of causal probes by replacing internal activations between “clean” and “corrupted” runs; however, methodological choices (metrics, corruption processes, and patch sites) materially affect conclusions, motivating explicit best-practice guidance [7]. Our benchmark is designed so that corruption and intervention targets correspond to known nuisance factors (e.g., rotation), allowing sharper interpretation of causal tests.

2.5 Architectures with built-in geometric structure

Interpretability interacts with inductive bias: if a model is constrained to respect known symmetries, the space of plausible internal mechanisms shrinks. Group-equivariant CNNs generalize standard convolutions to enforce equivariance under transformations beyond translation, including rotations and reflections [5]. Steerable CNNs provide a principled construction of equivariant convolutions for continuous groups such as $E(2)$, yielding structured feature fields with controlled transformation laws [13]. Scattering networks build invariant and deformation-stable representations using wavelet cascades with fixed filters and nonlinearities, offering an analytically tractable alternative that connects classical signal processing to deep convolutional ideas [3]. These lines of work inform our comparison between privileged-basis (fixed-template) layers and learned layers, especially when rotations are a dominant nuisance factor.

References

- [1] Bau, D., Zhou, B., Khosla, A., Oliva, A., and Torralba, A. (2017). Network dissection: Quantifying interpretability of deep visual representations. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3319–3327.
- [2] Box, G. E. P. (1976). Science and statistics. *Journal of the American Statistical Association*, 71(356):791–799.
- [3] Bruna, J. and Mallat, S. (2013). Invariant scattering convolution networks. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 35(8):1872–1886.

- [4] Chan, L., Garriga-Alonso, A., Goldowsky-Dill, N., Greenblatt, R., Nitishinskaya, J., Radhakrishnan, A., Shlegeris, B., and Thomas, N. (2022). Causal scrubbing, a method for rigorously testing interpretability hypotheses. *AI Alignment Forum*. <https://www.alignmentforum.org/posts/JvZhhzycHu2Yd57RN/causal-scrubbing-a-method-for-rigorously-testing>.
- [5] Cohen, T. and Welling, M. (2016). Group equivariant convolutional networks. In Balcan, M. F. and Weinberger, K. Q., editors, *Proceedings of The 33rd International Conference on Machine Learning*, volume 48 of *Proceedings of Machine Learning Research*, pages 2990–2999. PMLR.
- [6] Geiger, A., Ibeling, D., Zur, A., Chaudhary, M., Chauhan, S., Huang, J., Arora, A., Wu, Z., Goodman, N., Potts, C., and Icard, T. (2025). Causal abstraction: A theoretical foundation for mechanistic interpretability. *Journal of Machine Learning Research*, 26(83):1–64.
- [7] Heimersheim, S. and Nanda, N. (2024). How to use and interpret activation patching. *arXiv preprint arXiv:2404.15255*.
- [8] Kim, B., Wattenberg, M., Gilmer, J., Cai, C., Wexler, J., Viegas, F., and Sayres, R. (2018). Interpretability beyond feature attribution: Quantitative testing with concept activation vectors (tcav). In Dy, J. and Krause, A., editors, *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 2668–2677. PMLR.
- [9] Mahendran, A. and Vedaldi, A. (2015). Understanding deep image representations by inverting them. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5188–5196.
- [10] Olah, C., Mordvintsev, A., and Schubert, L. (2017). Feature visualization. *Distill*, 2(11).
- [11] Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., and Batra, D. (2017). Grad-cam: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 618–626.
- [12] Sundararajan, M., Taly, A., and Yan, Q. (2017). Axiomatic attribution for deep networks. In Precup, D. and Teh, Y. W., editors, *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 3319–3328. PMLR.
- [13] Weiler, M. and Cesa, G. (2019). General $e(2)$ -equivariant steerable cnns. In *Advances in Neural Information Processing Systems*, volume 32, pages 14334–14345.
- [14] Zeiler, M. D. and Fergus, R. (2014). Visualizing and understanding convolutional networks. In Fleet, D., Pajdla, T., Schiele, B., and Tuytelaars, T., editors, *Computer Vision – ECCV 2014*, volume 8689 of *Lecture Notes in Computer Science*, pages 818–833. Springer.